

Multivariate Methods

Métodos multivariados | Lecture notes

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1. Notation and conventions

These notes are bilingual: the derivations are written in English and the procedural summaries (*resumen del proceso*, *planteamiento*) are kept in Spanish, as in the original handwritten notes.

Throughout, $\mathbf{X}^\top = (X_1, X_2, \dots, X_p)$ is a random vector of p observed variables with mean vector $\boldsymbol{\mu} = \mathbb{E}(\mathbf{X})$ and covariance matrix

$$\Sigma = \text{Cov}(\mathbf{X}) = \mathbb{E}[(\mathbf{X} - \boldsymbol{\mu})(\mathbf{X} - \boldsymbol{\mu})^\top],$$

with diagonal entries $\Sigma_{ii} = \sigma_{ii} = \text{Var}(X_i)$.

- $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_p \geq 0$ are the eigenvalues of Σ (it is symmetric positive semidefinite, so they are real and nonnegative), with associated orthonormal eigenvectors $\mathbf{e}_1, \dots, \mathbf{e}_p$, i.e. $\mathbf{e}_i^\top \mathbf{e}_k = \delta_{ik}$.
- $\boldsymbol{\rho}$ denotes the correlation matrix and $\mathbf{Z}$ the vector of standardized variables.
- Transposition is written $\mathbf{a}^\top$; in the original notes this appears both as $\mathbf{a}'$ and as $\mathbf{Q}^T$, and both mean the same thing

here.

- m always denotes the number of common factors in Section 3, with $m < p$.
- G denotes the categorical class label and K the number of classes in Section 4; the handwritten notes use both N and K for this count, and here it is always K .
- In Section 5, $\sigma(\cdot)$ denotes the logistic function $\sigma(z) = 1/(1 + e^{-z})$ — not to be confused with the entries σ_{ii} of Σ — and $\ell(\beta)$ the log-likelihood. The handwritten notes write the logistic function as $\ln(z)$, which here is reserved for the natural logarithm.

2. Principal components

2.1. Setup: linear combinations of the observed variables

Let the random vector $\mathbf{X}^\top = (X_1, X_2, \dots, X_p)$ have covariance matrix Σ with eigenvalues $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_p \geq 0$. Consider the linear combinations

$$\begin{aligned}
Y_1 &= \mathbf{a}_1^\top \mathbf{X} = a_{11}X_1 + a_{12}X_2 + \cdots + a_{1p}X_p, \\
Y_2 &= \mathbf{a}_2^\top \mathbf{X} = a_{21}X_1 + a_{22}X_2 + \cdots + a_{2p}X_p, \\
&\vdots \\
Y_p &= \mathbf{a}_p^\top \mathbf{X} = a_{p1}X_1 + a_{p2}X_2 + \cdots + a_{pp}X_p.
\end{aligned} \tag{2.1.1}$$

Using the bilinearity of the covariance operator we obtain

$$\text{Var}(Y_i) = \mathbf{a}_i^\top \Sigma \mathbf{a}_i, \tag{2.1.2}$$

$$\text{Cov}(Y_i, Y_k) = \mathbf{a}_i^\top \Sigma \mathbf{a}_k, \tag{2.1.3}$$

for $i, k = 1, 2, \dots, p$.

Definition 2.1 (Principal components). The **principal components** are those uncorrelated linear combinations $Y_1, Y_2, \dots, Y_p$ of (2.1.1) whose variances (2.1.2) are as large as possible.

The first linear combination is the one of maximum variance, i.e. the one maximizing $\text{Var}(Y_1) = \mathbf{a}_1^\top \Sigma \mathbf{a}_1$. Since this variance can be made arbitrarily large by rescaling $\mathbf{a}_1$, the indeterminacy is eliminated by restricting attention to coefficient vectors of unit length:

$$\begin{aligned}
&\text{first principal component} = \mathbf{a}_1^\top \mathbf{X} \\
&\text{that maximizes } \text{Var}(\mathbf{a}_1^\top \mathbf{X}) \text{ s.t. } \mathbf{a}_1^\top \mathbf{a}_1 = 1.
\end{aligned} \tag{2.1.4}$$

In general, the i -th principal component is the combination $\mathbf{a}_i^\top \mathbf{X}$ that maximizes $\text{Var}(\mathbf{a}_i^\top \mathbf{X})$ subject to $\mathbf{a}_i^\top \mathbf{a}_i = 1$ and $\text{Cov}(\mathbf{a}_i^\top \mathbf{X}, \mathbf{a}_k^\top \mathbf{X}) = 0$ for all $k < i$.

2.2. The maximization lemma and the solution

The constrained problem (2.1.4) is solved by the maximization of a Rayleigh quotient.

Lemma 2.2 (Maximization of a quadratic form). Let $\mathbf{B} = \Sigma$ be positive definite with eigenvalue–eigenvector pairs $(\lambda_1, \mathbf{e}_1), \dots, (\lambda_p, \mathbf{e}_p)$ and $\lambda_1 \geq \dots \geq \lambda_p \geq 0$. Then

$$\max_{\mathbf{a} \neq \mathbf{0}} \frac{\mathbf{a}^\top \Sigma \mathbf{a}}{\mathbf{a}^\top \mathbf{a}} = \lambda_1, \tag{2.2.1}$$

attained at $\mathbf{a} = \mathbf{e}_1$, and, restricting to the orthogonal complement of the leading eigenvectors,

$$\max_{\mathbf{a} \perp \mathbf{e}_1, \dots, \mathbf{e}_k} \frac{\mathbf{a}^\top \Sigma \mathbf{a}}{\mathbf{a}^\top \mathbf{a}} = \lambda_{k+1}, \tag{2.2.2}$$

attained at $\mathbf{a} = \mathbf{e}_{k+1}$, for $k = 1, 2, \dots, p-1$.

Proof. Since the eigenvectors are normalized, $\mathbf{e}_1^\top \mathbf{e}_1 = 1$, and evaluating the quotient (2.2.1) at $\mathbf{a} = \mathbf{e}_1$ gives

$$\max_{\mathbf{a} \neq \mathbf{0}} \frac{\mathbf{a}^\top \Sigma \mathbf{a}}{\mathbf{a}^\top \mathbf{a}} = \lambda_1 = \frac{\mathbf{e}_1^\top \Sigma \mathbf{e}_1}{\mathbf{e}_1^\top \mathbf{e}_1} = \mathbf{e}_1^\top \Sigma \mathbf{e}_1 = \text{Var}(Y_1).$$

Similarly, for the choice $\mathbf{a} = \mathbf{e}_{k+1}$, which satisfies $\mathbf{e}_{k+1}^\top \mathbf{e}_i = 0$ for $i = 1, 2, \dots, k$ and $k = 1, 2, \dots, p-1$,

$$\frac{\mathbf{e}_{k+1}^\top \Sigma \mathbf{e}_{k+1}}{\mathbf{e}_{k+1}^\top \mathbf{e}_{k+1}} = \mathbf{e}_{k+1}^\top \Sigma \mathbf{e}_{k+1} = \text{Var}(Y_{k+1}),$$

and since $\Sigma \mathbf{e}_{k+1} = \lambda_{k+1} \mathbf{e}_{k+1}$,

$$\mathbf{e}_{k+1}^\top (\Sigma \mathbf{e}_{k+1}) = \lambda_{k+1} \mathbf{e}_{k+1}^\top \mathbf{e}_{k+1} = \lambda_{k+1},$$

so that $\text{Var}(Y_{k+1}) = \lambda_{k+1}$. $\square$

Theorem 2.3 (Principal components of Σ). Let Σ be the covariance matrix associated with the random vector $\mathbf{X}^\top = (X_1, X_2, \dots, X_p)$, and let Σ have the eigenvalue–eigenvector pairs $(\lambda_1, \mathbf{e}_1), (\lambda_2, \mathbf{e}_2), \dots, (\lambda_p, \mathbf{e}_p)$ with $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_p \geq 0$. Then the i -th principal component is given by

$$\begin{aligned}
Y_i &= \mathbf{e}_i^\top \mathbf{X} \\
&= e_{i1}X_1 + e_{i2}X_2 + \cdots + e_{ip}X_p,
\end{aligned} \tag{2.2.3}$$

for $i = 1, 2, \dots, p$, and with these choices

$$\text{Var}(Y_i) = \mathbf{e}_i^\top \Sigma \mathbf{e}_i = \lambda_i, \tag{2.2.4}$$

$$\text{Cov}(Y_i, Y_k) = \mathbf{e}_i^\top \Sigma \mathbf{e}_k = 0, \quad i \neq k. \tag{2.2.5}$$

Remark. If some λ_i are equal, the choices of the corresponding coefficient vectors $\mathbf{e}_i$ — and hence of Y_i — are not unique.

2.3. Uncorrelatedness of the components

It remains to show that the components are uncorrelated, that is, that $\mathbf{e}_i \perp \mathbf{e}_k \Rightarrow \mathbf{e}_i^\top \mathbf{e}_k = 0$ for $i \neq k$ gives $\text{Cov}(Y_i, Y_k) = 0$.

The eigenvectors of Σ are orthogonal if all the eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_p$ are distinct. If they are not all distinct, the eigenvectors corresponding to common eigenvalues may be chosen to be orthogonal. Therefore, for any two eigenvectors $\mathbf{e}_i$ and $\mathbf{e}_k$ we have $\mathbf{e}_i^\top \mathbf{e}_k = 0$ for $i \neq k$, and since $\Sigma \mathbf{e}_k = \lambda_k \mathbf{e}_k$, premultiplying by $\mathbf{e}_i^\top$ gives

$$\text{Cov}(Y_i, Y_k) = \mathbf{e}_i^\top \Sigma \mathbf{e}_k = \mathbf{e}_i^\top \lambda_k \mathbf{e}_k = \lambda_k \mathbf{e}_i^\top \mathbf{e}_k = 0, \tag{2.3.1}$$

for all $i \neq k$.

2.4. Total population variance

Theorem 2.4 (Decomposition of the total variance).

Let $\mathbf{X}^\top = (X_1, X_2, \dots, X_p)$ have covariance matrix Σ with pairs $(\lambda_1, \mathbf{e}_1), \dots, (\lambda_p, \mathbf{e}_p)$, where $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_p \geq 0$, and let $Y_1 = \mathbf{e}_1^\top \mathbf{X}, \dots, Y_p = \mathbf{e}_p^\top \mathbf{X}$ be the principal components. Then

$$\begin{aligned}
\sigma_{11} + \sigma_{22} + \cdots + \sigma_{pp} &= \sum_{i=1}^p \text{Var}(X_i) \\
&= \lambda_1 + \lambda_2 + \cdots + \lambda_p = \sum_{i=1}^p \text{Var}(Y_i).
\end{aligned} \tag{2.4.1}$$

Proof. From the definition of the trace, $\sigma_{11} + \sigma_{22} + \cdots + \sigma_{pp} = \text{tr}(\Sigma)$. With $\mathbf{A} = \Sigma$ we may write the spectral decomposition

$$\Sigma = \mathbf{P} \Lambda \mathbf{P}^\top,$$

where Λ is the diagonal matrix of eigenvalues and $\mathbf{P} = (\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_p)$ is orthogonal, so that $\mathbf{P}^\top \mathbf{P} = \mathbf{P} \mathbf{P}^\top = \mathbf{I}$. Then, by the cyclic property of the trace,

$$\text{tr}(\Sigma) = \text{tr}(\Lambda \mathbf{P}^\top \mathbf{P}) = \text{tr}(\Lambda) = \lambda_1 + \lambda_2 + \cdots + \lambda_p.$$

Thus

$$\sum_{i=1}^p \text{Var}(X_i) = \text{tr}(\Sigma) = \text{tr}(\Lambda) = \sum_{i=1}^p \text{Var}(Y_i),$$

because $\text{Var}(Y_i) = \lambda_i$ by Theorem 2.3. $\square$

Consequently the **total population variance** is

$$\sigma_{11} + \sigma_{22} + \cdots + \sigma_{pp} = \lambda_1 + \lambda_2 + \cdots + \lambda_p, \tag{2.4.2}$$

and the proportion of it accounted for by the k -th component is

$$\left(\begin{array}{c} \text{proportion of total} \\ \text{variance due to the} \\ k\text{-th component} \end{array} \right) = \frac{\lambda_k}{\lambda_1 + \lambda_2 + \dots + \lambda_p}, \quad (2.4.3)$$

for $k = 1, 2, \dots, p$.

2.5. Interpretation of the loadings

Consider each component of $\mathbf{e}_i = (e_{i1}, e_{i2}, \dots, e_{ip})$. The magnitude of e_{ik} measures the importance of the k -th variable to the i -th principal component; in particular e_{ik} is proportional to the correlation coefficient between Y_i and X_k ,

$$e_{ik} \propto \text{Corr}(Y_i, X_k).$$

Proposition 2.5 (Correlation of components and variables). *If $Y_1 = \mathbf{e}_1^T \mathbf{X}, \dots, Y_p = \mathbf{e}_p^T \mathbf{X}$ are the principal components obtained from Σ , then*

$$\rho_{Y_i, X_k} = \frac{e_{ik} \sqrt{\lambda_i}}{\sqrt{\sigma_{kk}}}, \quad i, k = 1, \dots, p. \quad (2.5.1)$$

2.6. Principal components from standardized variables

The standardized variables are

$$Z_1 = \frac{X_1 - \mu_1}{\sqrt{\sigma_{11}}}, \dots, Z_p = \frac{X_p - \mu_p}{\sqrt{\sigma_{pp}}}. \quad (2.6.1)$$

In matrix notation, where $\mathbf{V}^{1/2}$ is the diagonal standard deviation matrix,

$$\mathbf{Z} = \left(\mathbf{V}^{1/2} \right)^{-1} (\mathbf{X} - \boldsymbol{\mu}), \quad \mathbb{E}(\mathbf{Z}) = \mathbf{0}, \quad (2.6.2)$$

and

$$\text{Cov}(\mathbf{Z}) = \left(\mathbf{V}^{1/2} \right)^{-1} \Sigma \left(\mathbf{V}^{1/2} \right)^{-1} = \boldsymbol{\rho}. \quad (2.6.3)$$

The principal components may therefore be obtained from the eigenvectors of the *correlation matrix* $\boldsymbol{\rho}$ of $\mathbf{X}$. All the previous results apply, with some simplification given that the variance of each Z_i is unity.

Theorem 2.6 (Components of the standardized variables). *The i -th principal component of the standardized variables $\mathbf{Z}^T = (Z_1, Z_2, \dots, Z_p)$ with $\text{Cov}(\mathbf{Z}) = \boldsymbol{\rho}$ is given by*

$$Y_i = \mathbf{e}_i^T \mathbf{Z} = \mathbf{e}_i^T \left(\mathbf{V}^{1/2} \right)^{-1} (\mathbf{X} - \boldsymbol{\mu}), \quad (2.6.4)$$

where $(\lambda_1, \mathbf{e}_1), \dots, (\lambda_p, \mathbf{e}_p)$ are the *eigenvalue-eigenvector pairs* of $\boldsymbol{\rho}$ with $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_p \geq 0$. Moreover

$$\sum_{i=1}^p \text{Var}(Y_i) = \sum_{i=1}^p \text{Var}(Z_i) = p, \quad (2.6.5)$$

$$\rho_{Y_i, Z_k} = e_{ik} \sqrt{\lambda_i}. \quad (2.6.6)$$

We see from (2.6.5) that the total standardized population variance is simply p , the sum of the diagonal elements of the matrix $\boldsymbol{\rho}$. Working with $\mathbf{Z}$ in place of $\mathbf{X}$, we find the proportion of total variance explained by the k -th principal component of $\mathbf{Z}$ to be

$$\left(\begin{array}{c} \text{proportion of standardized} \\ \text{variance due to the} \\ k\text{-th component} \end{array} \right) = \frac{\lambda_k}{p}, \quad (2.6.7)$$

for $k = 1, 2, \dots, p$, where the λ_k are now the eigenvalues of $\boldsymbol{\rho}$.

2.7. Análisis de componentes principales

Sea Σ la matriz de covarianzas del conjunto de datos. La varianza total se da por la traza

$$\text{varianza total} = \text{tr } \Sigma = \sum_{i=1}^p \Sigma_{ii}, \quad (2.7.1)$$

donde Σ_{ii} son las varianzas individuales de cada variable.

Usando la descomposición en valores y vectores propios (SVD) de Σ , existen matrices $\mathbf{Q}$ y Λ tales que

$$\Sigma = \mathbf{Q} \Lambda \mathbf{Q}^T, \quad (2.7.2)$$

con Λ diagonal con valores propios $\lambda_1, \dots, \lambda_p$. La varianza total puede escribirse como

$$\begin{aligned} \text{varianza total} &= \text{tr } \Sigma = \text{tr } (\mathbf{Q} \Lambda \mathbf{Q}^T) \\ &= \text{tr } \Lambda = \lambda_1 + \lambda_2 + \dots + \lambda_p. \end{aligned} \quad (2.7.3)$$

Las columnas de $\mathbf{Q}$ son las direcciones de las componentes principales; los valores de Λ indican cuánta varianza explica cada componente. Estas componentes definen un nuevo sistema de coordenadas: cada punto se representa por sus puntuaciones de componentes, relacionadas con las variables originales por

$$\mathbf{X} = \mathbf{Q}(\text{PC}) + \bar{\mathbf{x}}, \quad (2.7.4)$$

donde $\mathbf{X}$ es el vector de coordenadas originales, PC el de puntuaciones y $\bar{\mathbf{x}}$ el vector de medias.

2.8. Resumen del proceso de PCA

Se resume según estos pasos:

1. **Estandarizar los datos.**
2. **Calcular la matriz de covarianzas.**
 - La matriz calcula cómo varían conjuntamente las variables.
 - Su tamaño es $p \times p$, donde p es el número de variables.
3. **Obtener la descomposición en valores y vectores propios (SVD).**
 - Los *eigenvalues* miden la varianza explicada por cada componente.
 - Los *eigenvectors* son las direcciones (combinaciones lineales) de las componentes.
4. **Ordenar las componentes.**
 - Ordenar valores propios de mayor a menor.
 - El vector correspondiente al mayor valor propio es la primera componente.
5. **Decidir cuántas componentes conservar.**
 - Criterio de Kaiser: conservar valores propios > 1 .
 - Gráfico *scree*: buscar el “codo”.
 - Varianza acumulada.
6. **Obtener puntuaciones:** proyectar los datos centrados sobre las componentes seleccionadas,

$$\text{PC} = \mathbf{Q}^T (\mathbf{X} - \bar{\mathbf{x}}). \quad (2.8.1)$$

3. Factor analysis

3.1. The orthogonal factor model

Let the random vector $\mathbf{X}$ with p components have mean $\boldsymbol{\mu}$ and covariance matrix $\mathbf{S}$ (equivalently Σ). The factor model postulates that the observed variables are linearly dependent on m unobservable random variables

$$F_1, F_2, \dots, F_m \quad (m < p),$$

called **common factors**, and on p additional sources of variation

$$\varepsilon_1, \varepsilon_2, \dots, \varepsilon_p,$$

called **errors** or **specific factors**. For p variables one looks for m factors (with $m < p$) that explain a high percentage of the variation and that have some meaning in the problem at hand. The model reads

$$\begin{aligned} X_1 - \mu_1 &= \ell_{11}F_1 + \ell_{12}F_2 + \dots + \ell_{1m}F_m + \varepsilon_1, \\ X_2 - \mu_2 &= \ell_{21}F_1 + \ell_{22}F_2 + \dots + \ell_{2m}F_m + \varepsilon_2, \\ &\vdots \\ X_p - \mu_p &= \ell_{p1}F_1 + \ell_{p2}F_2 + \dots + \ell_{pm}F_m + \varepsilon_p, \end{aligned} \quad (3.1.1)$$

or, in matrix notation,

$$\underset{(p \times 1)}{\mathbf{X}} - \underset{(p \times 1)}{\boldsymbol{\mu}} = \underset{(p \times m)}{\mathbf{L}} \underset{(m \times 1)}{\mathbf{F}} + \underset{(p \times 1)}{\boldsymbol{\varepsilon}}. \quad (3.1.2)$$

The coefficient ℓ_{ij} is the **loading** of the i -th variable on the j -th factor, so $\mathbf{L}$ is the matrix of factor loadings. The p deviations $X_i - \mu_i$ are expressed in terms of $p+m$ random variables $F_1, \dots, F_m, \varepsilon_1, \dots, \varepsilon_p$, which are unobservable. In summary:

- μ_i mean of variable i .
- ε_i i -th specific factor.
- F_j j -th common factor.
- ℓ_{ij} loading of the i -th variable on the j -th factor.

3.2. Covariance relationships implied by the model

The orthogonal factor model with m common factors,

$$\underset{(p \times 1)}{\mathbf{X}} = \underset{(p \times 1)}{\boldsymbol{\mu}} + \underset{(p \times m)}{\mathbf{L}} \underset{(m \times 1)}{\mathbf{F}} + \underset{(p \times 1)}{\boldsymbol{\varepsilon}}, \quad (3.2.1)$$

assumes that $\mathbf{F}$ and $\boldsymbol{\varepsilon}$ are independent, with

$$\mathbb{E}(\mathbf{F}) = \mathbf{0}, \quad \text{Cov}(\mathbf{F}) = \mathbb{E}[\mathbf{F}\mathbf{F}^\top] = \mathbf{I}_m, \quad (3.2.2)$$

$$\mathbb{E}(\boldsymbol{\varepsilon}) = \mathbf{0}, \quad \text{Cov}(\boldsymbol{\varepsilon}) = \mathbb{E}[\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}^\top] = \boldsymbol{\Psi}, \quad (3.2.3)$$

where $\boldsymbol{\Psi}$ is the diagonal matrix

$$\boldsymbol{\Psi} = \begin{pmatrix} \psi_1 & 0 & \dots & 0 \\ 0 & \psi_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \psi_p \end{pmatrix}, \quad (3.2.4)$$

and the factors are uncorrelated with the errors,

$$\text{Cov}(\boldsymbol{\varepsilon}, \mathbf{F}) = \mathbb{E}[\boldsymbol{\varepsilon}\mathbf{F}^\top] = \underset{(p \times m)}{\mathbf{0}}. \quad (3.2.5)$$

3.3. Planteamiento y consecuencias

El análisis factorial es una técnica estadística usada para identificar patrones ocultos o estructuras latentes detrás de grandes conjuntos de variables observadas. Busca reducir la complejidad agrupando variables relacionadas en un menor número de factores subyacentes más fáciles de interpretar.

El contexto es una relación de la forma

$$\mathbf{X} = \boldsymbol{\mu} + \mathbf{L}\mathbf{F} + \boldsymbol{\varepsilon}, \quad (3.3.1)$$

donde $\mathbf{X}$ es un vector aleatorio de p variables; $\boldsymbol{\mu}$ un vector constante de dimensión p ; $\mathbf{L}$ una matriz constante $p \times m$, la “matriz de cargas factoriales”; $\mathbf{F} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_m)$ el vector de m factores latentes, con media $\mathbf{0}$ y matriz de covarianzas identidad; y $\boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Psi})$ el vector de errores únicos, con distribución normal multivariante de vector de medias $\mathbf{0}$ y matriz de covarianzas diagonal $\boldsymbol{\Psi} = \text{diag}\{\sigma_1^2, \dots, \sigma_p^2\}$.

Se asume que los factores $\mathbf{F}$ y los errores $\boldsymbol{\varepsilon}$ son independientes, $\text{Cov}(\mathbf{F}, \boldsymbol{\varepsilon}) = \mathbf{0}$. Como consecuencia se tiene

$$\begin{aligned} \mathbb{E}(\mathbf{X}) &= \mathbb{E}(\boldsymbol{\mu} + \mathbf{L}\mathbf{F} + \boldsymbol{\varepsilon}) \\ &= \mathbb{E}(\boldsymbol{\mu}) + \mathbf{L}\mathbb{E}(\mathbf{F}) + \mathbb{E}(\boldsymbol{\varepsilon}) = \boldsymbol{\mu}, \end{aligned} \quad (3.3.2)$$

y para la varianza

$$\begin{aligned} \text{Var}(\mathbf{X}) &= \text{Var}(\mathbf{L}\mathbf{F} + \boldsymbol{\varepsilon}) \\ &= \mathbb{E}[(\mathbf{L}\mathbf{F} + \boldsymbol{\varepsilon})(\mathbf{L}\mathbf{F} + \boldsymbol{\varepsilon})^\top] \\ &= \mathbb{E}[\mathbf{L}\mathbf{F}\mathbf{F}^\top\mathbf{L}^\top] + \mathbb{E}[\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}^\top] \\ &\quad + \mathbb{E}[\mathbf{L}\mathbf{F}\boldsymbol{\varepsilon}^\top] + \mathbb{E}[\boldsymbol{\varepsilon}\mathbf{F}^\top\mathbf{L}^\top] \\ &= \mathbf{L}\mathbb{E}[\mathbf{F}\mathbf{F}^\top]\mathbf{L}^\top + \mathbb{E}[\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}^\top] \\ &\quad + \mathbf{L}\mathbb{E}[\mathbf{F}\boldsymbol{\varepsilon}^\top] + \mathbb{E}[\boldsymbol{\varepsilon}\mathbf{F}^\top]\mathbf{L}^\top \\ &= \mathbf{L}\mathbf{I}_m\mathbf{L}^\top + \boldsymbol{\Psi} + \mathbf{L}\mathbf{0} + \mathbf{0}\mathbf{L}^\top \\ &= \mathbf{L}\mathbf{L}^\top + \boldsymbol{\Psi}. \end{aligned} \quad (3.3.3)$$

Como $\mathbf{X}$ es combinación lineal de normales, se sigue que

$$\mathbf{X} \sim \mathcal{N}(\boldsymbol{\mu}, \mathbf{L}\mathbf{L}^\top + \boldsymbol{\Psi}). \quad (3.3.4)$$

3.4. Descomposición de la varianza

La descomposición $\text{Var}(\mathbf{X}) = \mathbf{L}\mathbf{L}^\top + \boldsymbol{\Psi}$ muestra que la varianza total de las variables observadas puede repartirse en dos componentes:

- **Varianza común** ($\mathbf{L}\mathbf{L}^\top$): explicada por los factores comunes $\mathbf{F}$; capta la variabilidad compartida atribuible a la estructura latente.
- **Varianza única** ($\boldsymbol{\Psi}$): específica de cada variable observada.

La varianza de cada variable observada X_i se expresa como

$$\text{Var}(X_i) = \sum_{j=1}^m \ell_{ij}^2 + \sigma_i^2, \quad (3.4.1)$$

donde

$$h_i^2 = \sum_{j=1}^m \ell_{ij}^2 \quad (3.4.2)$$

es la **comunalidad** de X_i (la parte explicable por los factores comunes) y σ_i^2 la **varianza única**.

3.5. Estimación de puntuaciones factoriales

Supongamos $\mathbf{X} = \boldsymbol{\mu} + \mathbf{L}\mathbf{F} + \boldsymbol{\varepsilon}$. La distribución conjunta de $\mathbf{X}$ y $\mathbf{F}$ es

$$\begin{bmatrix} \mathbf{X} \\ \mathbf{F} \end{bmatrix} \sim \mathcal{N}\left(\begin{bmatrix} \boldsymbol{\mu} \\ \mathbf{0} \end{bmatrix}, \begin{bmatrix} \mathbf{L}\mathbf{L}^\top + \boldsymbol{\Psi} & \mathbf{L} \\ \mathbf{L}^\top & \mathbf{I}_m \end{bmatrix}\right). \quad (3.5.1)$$

Nótese en particular que

$$\begin{aligned} \text{Cov}(\mathbf{X}, \mathbf{F}) &= \mathbb{E}[(\mathbf{X} - \boldsymbol{\mu})(\mathbf{F} - \mathbf{0})^\top] \\ &= \mathbb{E}[(\mathbf{L}\mathbf{F} + \boldsymbol{\varepsilon})\mathbf{F}^\top] \\ &= \mathbf{L}\mathbb{E}[\mathbf{F}\mathbf{F}^\top] + \mathbb{E}[\boldsymbol{\varepsilon}\mathbf{F}^\top] \\ &= \mathbf{L}\mathbf{I}_m + \mathbf{0} = \mathbf{L}. \end{aligned} \quad (3.5.2)$$

Usando las propiedades de la normal multivariante, se obtiene la distribución condicional de $\mathbf{F}$ dado $\mathbf{X}$:

$$\begin{aligned} [\mathbf{F} | \mathbf{X} = \mathbf{x}] &\sim \mathcal{N}\left(\mathbf{L}^\top (\mathbf{L}\mathbf{L}^\top + \boldsymbol{\Psi})^{-1} (\mathbf{x} - \boldsymbol{\mu}), \right. \\ &\quad \left. \mathbf{I}_m - \mathbf{L}^\top (\mathbf{L}\mathbf{L}^\top + \boldsymbol{\Psi})^{-1} \mathbf{L}\right). \end{aligned} \quad (3.5.3)$$

La media de esta distribución se conoce como la **puntuación factorial** para $\mathbf{X} = \mathbf{x}$:

$$\hat{\mathbf{F}} = \mathbf{L}^\top (\mathbf{L}\mathbf{L}^\top + \mathbf{\Psi})^{-1} (\mathbf{x} - \boldsymbol{\mu}). \quad (3.5.4)$$

4. Discriminant analysis

4.1. The Gaussian discriminant model

Suppose the population is partitioned into K distinct classes by a categorical variable $G \in \{1, 2, \dots, K\}$, and let

$$\mathbf{X}^\top = (X_1, X_2, \dots, X_p)$$

be a continuous random vector of p observed variables. The **Gaussian discriminant model** (GDM) assumes that, within each class, $\mathbf{X}$ follows a multivariate normal distribution,

$$\mathbf{X} \mid (G = i) \sim \mathcal{N}(\boldsymbol{\mu}_i, \Sigma), \quad i = 1, \dots, K, \quad (4.1.1)$$

where the mean vector $\boldsymbol{\mu}_i$ depends on the class but the covariance matrix Σ is *the same* for every group. This last assumption is called **homoscedasticity**, and it is what makes the resulting rule linear.

Definition 4.1 (Classification function). A **classification function** is a map g that assigns to each observed value $\mathbf{X} = \mathbf{x}$ a class $G = i$, i.e. $g(\mathbf{x}) = i$.

4.2. The Bayes classifier

The goal is to find a g minimizing the **misclassification rate**

$$R(g) = \mathbb{E} [\mathbf{1}_{\{g(\mathbf{x}) \neq G\}}] = \mathbb{P}(g(\mathbf{X}) \neq G), \quad (4.2.1)$$

where the indicator takes the value

$$\mathbf{1}_{\{g(\mathbf{x}) \neq G\}} = \begin{cases} 1, & \text{incorrect classification,} \\ 0, & \text{correct classification.} \end{cases} \quad (4.2.2)$$

The risk (4.2.1) is minimized by the **Bayes classifier**, which assigns $\mathbf{x}$ to the class with the largest posterior probability:

$$\tilde{g}(\mathbf{x}) = \arg \max_i \mathbb{P}(G = i \mid \mathbf{X} = \mathbf{x}), \quad (4.2.3)$$

where $\mathbb{P}(G = i \mid \mathbf{X} = \mathbf{x})$ is the probability that an individual belongs to class i given that its observed characteristics are $\mathbf{X} = \mathbf{x}$. It remains to make (4.2.3) explicit under the GDM.

4.3. Explicit form of the classifier under the GDM

Introduce the **prior probabilities** of each class,

$$\pi_i = \mathbb{P}(G = i), \quad \sum_{i=1}^K \pi_i = 1, \quad (4.3.1)$$

and recall that under (4.1.1) the class-conditional density of $\mathbf{X}$ is

$$f_i(\mathbf{x}) = \frac{1}{(2\pi)^{p/2} |\Sigma|^{1/2}} \exp \left(-\frac{1}{2} (\mathbf{x} - \boldsymbol{\mu}_i)^\top \Sigma^{-1} (\mathbf{x} - \boldsymbol{\mu}_i) \right). \quad (4.3.2)$$

By Bayes' theorem the posterior probability is

$$\mathbb{P}(G = i \mid \mathbf{X} = \mathbf{x}) = \frac{\pi_i f_i(\mathbf{x})}{\sum_{j=1}^K \pi_j f_j(\mathbf{x})}. \quad (4.3.3)$$

Since the denominator of (4.3.3) is the same for every class, it does not affect the arg max, so

$$\mathbb{P}(G = i \mid \mathbf{X} = \mathbf{x}) \propto \pi_i f_i(\mathbf{x}), \quad (4.3.4)$$

and it suffices to concentrate on $\pi_i f_i(\mathbf{x})$. Taking logarithms (a monotone transformation, so the arg max is unchanged),

$$\begin{aligned} \log(\pi_i f_i(\mathbf{x})) &= \log \pi_i + \log f_i(\mathbf{x}) \\ &= \log \pi_i - \frac{p}{2} \log(2\pi) - \frac{1}{2} \log |\Sigma| \\ &\quad - \frac{1}{2} (\mathbf{x} - \boldsymbol{\mu}_i)^\top \Sigma^{-1} (\mathbf{x} - \boldsymbol{\mu}_i) \\ &= \underbrace{\log \pi_i - \frac{p}{2} \log(2\pi) - \frac{1}{2} \log |\Sigma| - \frac{1}{2} \mathbf{x}^\top \Sigma^{-1} \mathbf{x}}_{\text{terms common to all classes}} \\ &\quad + \boldsymbol{\mu}_i^\top \Sigma^{-1} \mathbf{x} - \frac{1}{2} \boldsymbol{\mu}_i^\top \Sigma^{-1} \boldsymbol{\mu}_i. \end{aligned} \quad (4.3.5)$$

Remark. The quadratic term $-\frac{1}{2} \mathbf{x}^\top \Sigma^{-1} \mathbf{x}$ survives the expansion of $(\mathbf{x} - \boldsymbol{\mu}_i)^\top \Sigma^{-1} (\mathbf{x} - \boldsymbol{\mu}_i)$ but does *not* depend on i precisely because Σ is common to all groups. This is where homoscedasticity is used, and it is why the rule is linear in $\mathbf{x}$; if each class had its own Σ_i the term would remain and one would obtain *quadratic* discriminant analysis instead.

Discarding the class-independent terms of (4.3.5) leads to the following definition.

Definition 4.2 (Linear discriminant functions). For each class $i = 1, \dots, K$,

$$\delta_i(\mathbf{x}) = \boldsymbol{\mu}_i^\top \Sigma^{-1} \mathbf{x} - \frac{1}{2} \boldsymbol{\mu}_i^\top \Sigma^{-1} \boldsymbol{\mu}_i + \log \pi_i. \quad (4.3.6)$$

Their pairwise differences are

$$\begin{aligned} \Delta_{ij}(\mathbf{x}) &= \log(\pi_i f_i(\mathbf{x})) - \log(\pi_j f_j(\mathbf{x})) \\ &= \delta_i(\mathbf{x}) - \delta_j(\mathbf{x}), \end{aligned} \quad (4.3.7)$$

which are affine functions of $\mathbf{x}$; the set $\{\Delta_{ij}(\mathbf{x}) = 0\}$ is therefore a hyperplane, the decision boundary between classes i and j . Finally, the Bayes classifier (4.2.3) takes the explicit form

$$\tilde{g}(\mathbf{x}) = \arg \max_i \delta_i(\mathbf{x}). \quad (4.3.8)$$

4.4. Canonical variables and the decomposition of variance

Discriminant analysis also allows one to define **canonical variables**: linear combinations that retain the information needed to separate the classes while reducing the dimension. The total mean is

$$\boldsymbol{\mu} = \mathbb{E}(\mathbf{X}) = \sum_{i=1}^K \pi_i \boldsymbol{\mu}_i, \quad (4.4.1)$$

and one introduces three dispersion matrices.

Definition 4.3 (Dispersion matrices). The **total** dispersion matrix is

$$\mathbf{S}_T = \mathbb{E}[(\mathbf{X} - \boldsymbol{\mu})(\mathbf{X} - \boldsymbol{\mu})^\top] = \text{Var}(\mathbf{X}), \quad (4.4.2)$$

the **within-groups** dispersion matrix is

$$\mathbf{S}_W = \mathbb{E}_G[\text{Var}(\mathbf{X} \mid G)] = \sum_{i=1}^K \pi_i \Sigma = \Sigma, \quad (4.4.3)$$

and the **between-groups** dispersion matrix is

$$\mathbf{S}_B = \text{Var}_G [\mathbb{E}(\mathbf{X} | G)] = \sum_{i=1}^K \pi_i (\boldsymbol{\mu}_i - \boldsymbol{\mu})(\boldsymbol{\mu}_i - \boldsymbol{\mu})^\top, \quad (4.4.4)$$

where $\mathbb{E}_G$ denotes expectation with respect to G and $\text{Var}(\mathbf{X} | G)$ the conditional variance of $\mathbf{X}$ given G .

Note that (4.4.3) collapses to Σ itself, again by homoscedasticity. The law of total variance then gives the **decomposition of variance**

$$\mathbf{S}_T = \mathbf{S}_W + \mathbf{S}_B. \quad (4.4.5)$$

Let now $\mathbf{a} = (a_1, a_2, \dots, a_p)$ be a $1 \times p$ matrix and consider the random variable $Y = \mathbf{a}\mathbf{X}$, a linear combination of the original variables $X_1, X_2, \dots, X_p$. Its total variance is

$$\text{Var}(Y) = \mathbf{a} \mathbf{S}_T \mathbf{a}^\top, \quad (4.4.6)$$

while its variance within the groups is

$$\mathbb{E}_G [\text{Var}(Y | G)] = \mathbf{a} \mathbf{S}_W \mathbf{a}^\top = \mathbf{a} \Sigma \mathbf{a}^\top. \quad (4.4.7)$$

To construct canonical variables one maximizes the **Rayleigh quotient**

$$J(\mathbf{a}) = \frac{\mathbf{a} \mathbf{S}_B \mathbf{a}^\top}{\mathbf{a} \mathbf{S}_W \mathbf{a}^\top}, \quad (4.4.8)$$

i.e. one seeks the direction along which the between-group spread is largest relative to the within-group spread.

4.5. Fisher's criterion

Fisher's criterion states that the canonical variables must satisfy:

- $Y_1 = \mathbf{a}_1 \mathbf{X}$ maximizes $J(\mathbf{a})$, that is,

$$\mathbf{a}_1 = \arg \max_{\mathbf{a}} J(\mathbf{a}); \quad (4.5.1)$$

- $Y_j = \mathbf{a}_j \mathbf{X}$ maximizes $J(\mathbf{a})$ subject to being unrelated to the previously obtained variables,

$$\mathbf{a}_j = \arg \max_{\mathbf{a}} J(\mathbf{a}) \quad \text{s.t.} \quad \text{Cov}(Y_i, Y_j | G) = 0, \quad i < j. \quad (4.5.2)$$

The maximum number of canonical variables is $\min(p, K-1)$, since by (4.4.4) the matrix $\mathbf{S}_B$ is a sum of K rank-one terms whose centred means satisfy one linear relation, so

$$r = \text{rank}(\mathbf{S}_B) \leq \min(p, K-1). \quad (4.5.3)$$

One verifies that the differences $\Delta_{ij}(\mathbf{x}) = \delta_i(\mathbf{x}) - \delta_j(\mathbf{x})$ of (4.3.7) depend on $\mathbf{x}$ only through the r canonical variables; the same holds for the Bayes classifier (4.3.8). Hence classification can be carried out in the reduced space of dimension $r \leq \min(p, K-1)$ without any loss of discriminating information.

4.6. Estimating the model from labelled data

Everything above is stated at the population level, in terms of the unknown parameters $\boldsymbol{\mu}_i$, Σ and π_i . In practice they must be estimated from a labelled sample

$$\{(\mathbf{x}_n, g_n)\}_{n=1}^m, \quad (4.6.1)$$

where $\mathbf{x}_n$ is the vector of observed characteristics of the n -th individual and $g_n \in \{1, \dots, K\}$ its known class. Write

$$m_i = \#\{n | g_n = i\}, \quad \sum_{i=1}^K m_i = m, \quad (4.6.2)$$

for the number of observations falling in class i . The class means are estimated by their sample counterparts,

$$\hat{\boldsymbol{\mu}}_i = \frac{1}{m_i} \sum_{n: g_n=i} \mathbf{x}_n. \quad (4.6.3)$$

For the common covariance matrix Σ there are two competing approaches: the **FE** estimator, which uses the unbiased pooled sum of squares, and the one based on **maximum likelihood** (MLE).

Definition 4.4 (Pooled covariance estimators).

$$\hat{\Sigma}_{\text{FE}} = \frac{1}{m-K} \sum_{i=1}^K \sum_{n: g_n=i} (\mathbf{x}_n - \hat{\boldsymbol{\mu}}_i)(\mathbf{x}_n - \hat{\boldsymbol{\mu}}_i)^\top, \quad (4.6.4)$$

$$\hat{\Sigma}_{\text{MLE}} = \frac{1}{m} \sum_{i=1}^K \sum_{n: g_n=i} (\mathbf{x}_n - \hat{\boldsymbol{\mu}}_i)(\mathbf{x}_n - \hat{\boldsymbol{\mu}}_i)^\top. \quad (4.6.5)$$

The two differ only in the normalising constant, so they are proportional:

$$\hat{\Sigma}_{\text{FE}} = \frac{m}{m-K} \hat{\Sigma}_{\text{MLE}}. \quad (4.6.6)$$

Remark. Estimating one mean vector per class consumes K degrees of freedom, so (4.6.4) divides by $m-K$ and is unbiased for Σ , whereas (4.6.5) divides by m and underestimates the dispersion. By (4.6.6) the correction factor $m/(m-K) \rightarrow 1$ as m grows, so the choice only matters when the sample size is close to the number of classes.

Finally the prior probabilities (4.3.1) are estimated by the observed class frequencies,

$$\hat{\pi}_i = \frac{m_i}{m}, \quad (4.6.7)$$

unless reliable external knowledge fixes them at other values. Substituting $\hat{\boldsymbol{\mu}}_i$, $\hat{\Sigma}$ and $\hat{\pi}_i$ into (4.3.6) gives the sample discriminant functions, and the classification rule (4.3.8) is applied with them.

Remark. Both approaches are implemented in R through the function `lda()` of the package `MASS`.

4.7. Checking the assumptions

The GDM rests on the two hypotheses of (4.1.1), and both should be checked before the rule is used.

- **Within-class multivariate normality.** The observations of each class must follow a multivariate normal distribution. This can be assessed with formal tests such as those of *Mardia*, *Henze-Zirkler* or *Royston*, and graphically with a multivariate Q-Q plot or with pairwise scatter plots.
- **Homoscedasticity.** The equality of the covariance matrices across classes is evaluated with *Box's M* test, available in R as `boxM()` in the package `biotools`. Scatter plots are again a useful complement to the formal test.

If homoscedasticity is rejected, the quadratic term of (4.3.5) no longer cancels and one is led to quadratic discriminant analysis instead.

4.8. Variable selection

Discarding predictors that carry no discriminating information simplifies the model, removes noise and makes the result easier to interpret. Let μ_{ij} denote the j -th component of $\boldsymbol{\mu}_i$, that is, the mean of the predictor X_j in class i . The variable

X_j is worth keeping only if that mean varies across the classes, which is exactly the hypothesis tested by a one-way ANOVA of X_j on the categorical variable G :

$$\begin{aligned} H_0 : \mu_{1j} &= \mu_{2j} = \dots = \mu_{Kj}, \\ H_1 : &\text{at least one of the means differs.} \end{aligned} \quad (4.8.1)$$

Fix a significance level α . For each variable X_j , $j = 1, \dots, p$:

1. perform a one-way ANOVA of X_j on G ;
2. obtain the corresponding p -value p_j ;
3. if $p_j \leq \alpha$, retain X_j ;
4. if $p_j > \alpha$, discard X_j .

4.9. Resumen del proceso

El análisis discriminante busca una regla que asigne cada individuo a una de K clases a partir de sus p características observadas. El modelo discriminante gaussiano supone normalidad dentro de cada clase y *homocedasticidad*, $\mathbf{X} \mid (G = i) \sim \mathcal{N}(\boldsymbol{\mu}_i, \Sigma)$ con Σ común a todos los grupos.

1. Se fija el objetivo: minimizar la tasa de error de clasificación $R(g) = \mathbb{P}(g(\mathbf{X}) \neq G)$. El óptimo es el clasificador de Bayes, que asigna $\mathbf{x}$ a la clase de mayor probabilidad a posteriori.
2. Se introducen las probabilidades a priori $\pi_i = \mathbb{P}(G = i)$ y se aplica el teorema de Bayes; el denominador es común a todas las clases, de modo que basta comparar $\pi_i f_i(\mathbf{x})$.
3. Al tomar logaritmos, los términos $-\frac{p}{2} \log(2\pi)$, $-\frac{1}{2} \log|\Sigma|$ y $-\frac{1}{2} \mathbf{x}^T \Sigma^{-1} \mathbf{x}$ aparecen en todas las clases y se descartan. Lo que queda son las **funciones discriminantes lineales** $\delta_i(\mathbf{x})$ de (4.3.6), y el clasificador es $\tilde{g}(\mathbf{x}) = \arg \max_i \delta_i(\mathbf{x})$.
4. Para reducir dimensión se descompone la varianza, $\mathbf{S}_T = \mathbf{S}_W + \mathbf{S}_B$, y se maximiza el cociente de Rayleigh $J(\mathbf{a}) = \mathbf{a} \mathbf{S}_B \mathbf{a}^T / \mathbf{a} \mathbf{S}_W \mathbf{a}^T$.
5. El criterio de Fisher genera así a lo más $\min(p, K - 1)$ variables canónicas, incorreladas dentro de los grupos, que contienen toda la información discriminante.

4.10. Elección del estimador de Σ

Los estimadores (4.6.4) y (4.6.5) sólo difieren en la constante de normalización, pero esa diferencia se vuelve relevante según el contexto:

Contexto	FE	MLE
Tamaño de muestra muy pequeño	✓	
Priors muy desbalanceados	✓	
Confianza alta en los valores de los priors	✓	
Clases balanceadas		✓
Clases bien separadas		✓
Alta dimensionalidad	✓	
Muestras muy grandes	✓	✓

4.11. El flujo del modelo LDA

1. Partición de los datos en entrenamiento y prueba.
2. Verificación de los supuestos (normalidad multivariada y homocedasticidad).
3. Selección de variables mediante ANOVA.
4. Construcción del modelo.
5. Validación cruzada.
6. Ajuste del modelo final.
7. Evaluación final sobre los datos de prueba.

5. Logistic regression

The logistic regression model predicts the probability that an observation belongs to a class. Unlike the Gaussian discriminant model of Section 4, which specifies the distribution of $\mathbf{X}$ within each class and then inverts it with Bayes' theorem,

here the posterior probability itself is modelled directly as a function of the predictors.

5.1. The logistic function

Definition 5.1 (Logistic function). The **logistic function** is

$$p = \sigma(z) = \frac{1}{1 + e^{-z}}, \quad z \in \mathbb{R}. \quad (5.1.1)$$

It is strictly increasing, satisfies $\sigma(z) \rightarrow 0$ as $z \rightarrow -\infty$ and $\sigma(z) \rightarrow 1$ as $z \rightarrow +\infty$, and $\sigma(0) = \frac{1}{2}$; hence it is a bijection

$$\mathbb{R} \longleftrightarrow (0, 1), \quad (5.1.2)$$

which is precisely what is needed to turn an unbounded linear predictor into a probability.

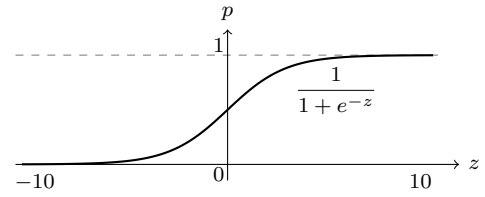


Figure 1: The logistic curve maps $\mathbb{R}$ onto $(0, 1)$.

5.2. The logit transformation

Solving (5.1.1) for z in terms of p gives

$$\begin{aligned} p &= \frac{1}{1 + e^{-z}} \Rightarrow p(1 + e^{-z}) = 1 \Rightarrow 1 + e^{-z} = \frac{1}{p} \\ &\Rightarrow e^{-z} = \frac{1}{p} - 1 = \frac{1 - p}{p} \\ &\Rightarrow -z = \log\left(\frac{1 - p}{p}\right) \\ &\Rightarrow z = \log\left(\frac{p}{1 - p}\right). \end{aligned} \quad (5.2.1)$$

Definition 5.2 (Logit function). The inverse of (5.1.1) is the **logit function**

$$z = \text{logit}(p) = \log\left(\frac{p}{1 - p}\right), \quad p \in (0, 1), \quad (5.2.2)$$

which transforms probabilities into real values.

The argument of (5.2.2) is itself worth naming.

Definition 5.3 (Odds). The **odds** (*oportunidad a momio*) associated with a probability p is

$$\frac{p}{1 - p} \in (0, \infty), \quad (5.2.3)$$

the ratio between the probability of the event and that of its complement.

So the logit is the logarithm of the odds, and (5.1.1) and (5.2.2) are inverse to each other.

5.3. The model

Suppose the response variable Y_0 takes the values 0 and 1, and that a vector of predictors

$$\mathbf{X}_0^T = (X_{01}, X_{02}, \dots, X_{0p}) \quad (5.3.1)$$

is observed. Write

$$p = \mathbb{P}(Y_0 = 1 \mid \mathbf{X}_0 = \mathbf{x}_0), \quad (5.3.2)$$

so that, conditionally on the predictors,

$$Y_0 \mid (\mathbf{X}_0 = \mathbf{x}_0) \sim \text{Bernoulli}(p). \quad (5.3.3)$$

Definition 5.4 (Logistic regression model). The **logistic regression model** assumes that the logit of p is an affine function of the predictors,

$$z = \log\left(\frac{p}{1-p}\right) = \beta_0 + \beta_1 x_{01} + \cdots + \beta_p x_{0p}. \quad (5.3.4)$$

Inverting (5.3.4) with (5.1.1) recovers the probability,

$$p = \sigma(z) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_{01} + \cdots + \beta_p x_{0p})}}, \quad (5.3.5)$$

and exponentiating (5.3.4) gives the odds

$$\frac{p}{1-p} = e^{(\beta_0 + \beta_1 x_{01} + \cdots + \beta_p x_{0p})}. \quad (5.3.6)$$

Remark. Equation (5.3.6) is what makes the coefficients interpretable. Increasing x_{0j} by one unit, with all other predictors held fixed, multiplies the odds by e^{β_j} ; the effect is additive on the log-odds scale and multiplicative on the odds scale, but *not* linear on the probability scale.

5.4. The logistic classifier

The model returns a probability, not a class. To obtain a classification function in the sense of Section 4 one fixes a **threshold** $\tau \in (0, 1)$ and assigns

$$g(\mathbf{x}_0) = \begin{cases} 1, & p \geq \tau, \\ 0, & p < \tau. \end{cases} \quad (5.4.1)$$

The **decision boundary** is the set where the two classes are declared equally plausible, that is, where $p = \tau$. Applying the logit to both sides and using (5.3.4),

$$\log\left(\frac{\tau}{1-\tau}\right) = \beta_0 + \beta_1 x_{01} + \cdots + \beta_p x_{0p}. \quad (5.4.2)$$

In particular, for the symmetric threshold $\tau = \frac{1}{2}$ the left-hand side vanishes and the boundary is

$$0 = \beta_0 + \beta_1 x_{01} + \cdots + \beta_p x_{0p}. \quad (5.4.3)$$

Remark. Equation (5.4.2) is affine in $\mathbf{x}_0$, so the boundary is a hyperplane, exactly as in (4.3.7): changing τ only shifts the intercept, never the orientation. Logistic regression and LDA therefore produce boundaries of the same shape; what differs is how the coefficients are obtained — LDA estimates μ_i and Σ and derives them, whereas logistic regression fits them directly.

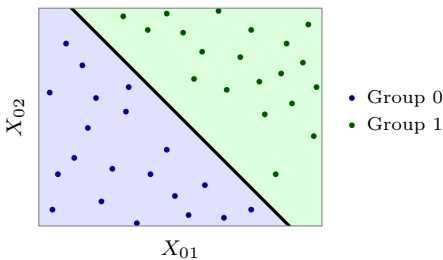


Figure 2: Example of the logistic classifier: the decision boundary (5.4.3) separates the two groups by a straight line in the plane of the predictors.

5.5. Estimating the coefficients

The coefficients $\beta^\top = (\beta_0, \beta_1, \dots, \beta_p)$ are unknown and must be estimated from a labelled sample $\{(\mathbf{x}_n, y_n)\}_{n=1}^m$ with $y_n \in \{0, 1\}$. Write

$$p_n = \sigma\left(\beta_0 + \sum_{j=1}^p \beta_j x_{nj}\right) \quad (5.5.1)$$

for the probability the model assigns to the n -th observation. Because the observations are independent Bernoulli variables (5.3.3), the likelihood is

$$L(\beta) = \prod_{n=1}^m p_n^{y_n} (1 - p_n)^{1-y_n}, \quad (5.5.2)$$

and its logarithm, the **log-likelihood**, is

$$\ell(\beta) = \sum_{n=1}^m [y_n \log p_n + (1 - y_n) \log (1 - p_n)]. \quad (5.5.3)$$

Definition 5.5 (Maximum likelihood estimator). The estimator $\hat{\beta}$ is the maximiser of (5.5.3),

$$\hat{\beta} = \arg \max_{\beta} \ell(\beta). \quad (5.5.4)$$

Collecting the predictors in the $m \times (p + 1)$ design matrix $\mathbf{X}$ whose first column is a column of ones, and writing $\mathbf{p} = (p_1, \dots, p_m)^\top$, the score equations obtained from (5.5.3) are

$$\frac{\partial \ell}{\partial \beta} = \mathbf{X}^\top (\mathbf{y} - \mathbf{p}) = \mathbf{0}. \quad (5.5.5)$$

Remark. System (5.5.5) is nonlinear in β , because $\mathbf{p}$ depends on β through the logistic function, and admits no closed-form solution — in contrast with ordinary least squares. It is solved numerically by Newton–Raphson, which in this setting takes the form of *iteratively reweighted least squares*,

$$\beta^{(t+1)} = \beta^{(t)} + (\mathbf{X}^\top \mathbf{W} \mathbf{X})^{-1} \mathbf{X}^\top (\mathbf{y} - \mathbf{p}), \quad (5.5.6)$$

with $\mathbf{W} = \text{diag}\{p_n(1 - p_n)\}$. Since ℓ is concave, the maximum is unique whenever the classes are not perfectly separable.

Remark. In R the model is fitted with `glm(y ~ x, family = binomial)`.

5.6. Adequacy of the model

To decide whether a subset of the predictors can be dropped one compares a **reduced** model, with q of the coefficients set to zero, against the **full** model, by means of the **likelihood ratio test**. The hypotheses are

$$\begin{aligned} H_0 : \beta_{j_1} = \cdots = \beta_{j_q} &= 0, \\ H_1 : \text{at least one of them is nonzero,} \end{aligned} \quad (5.6.1)$$

and the statistic is

$$\Lambda = -2 \left[\ell(\hat{\beta}_{\text{red}}) - \ell(\hat{\beta}_{\text{full}}) \right], \quad (5.6.2)$$

which under H_0 is asymptotically distributed as χ_q^2 . Large values of (5.6.2) indicate that the reduced model fits significantly worse, so H_0 is rejected and the predictors are kept.

Remark. Taking as reduced model the one with only the intercept ($\beta_1 = \cdots = \beta_p = 0$) turns (5.6.2) into a global test of the whole regression, the analogue of the overall F test in linear regression.

5.7. Resumen del proceso

La regresión logística predice la probabilidad de que una observación pertenezca a una clase, modelando directamente $p = \mathbb{P}(Y_0 = 1 \mid \mathbf{X}_0 = \mathbf{x}_0)$ en lugar de la distribución de las predictoras dentro de cada grupo.

1. La función logística $\sigma(z) = 1/(1 + e^{-z})$ transforma la recta real en el intervalo $(0, 1)$, de modo que su salida puede leerse como una probabilidad.
2. Su inversa es la función **logit**, $z = \log(p/(1 - p))$, que transforma probabilidades en valores reales; el cociente $p/(1 - p)$ es la *oportunidad a momio* (*odds*).
3. El modelo supone $Y_0 \mid (\mathbf{X}_0 = \mathbf{x}_0) \sim \text{Bernoulli}(p)$ con $\text{logit}(p) = \beta_0 + \beta_1 x_{01} + \cdots + \beta_p x_{0p}$; al despejar, $p = \sigma(z)$ y los momios quedan como en (5.3.6). Por eso e^{β_j} es el factor por el que se multiplican los momios al aumentar x_{0j} en una unidad.
4. Para clasificar se fija un umbral $\tau \in (0, 1)$; la frontera de decisión está dada por $\log(\tau/(1 - \tau)) = \beta_0 + \cdots + \beta_p x_{0p}$, que es un hiperplano. Con $\tau = 1/2$ la frontera es $0 = \beta_0 + \cdots + \beta_p x_{0p}$.
5. Los coeficientes se estiman por **máxima verosimilitud**; las ecuaciones de score $\mathbf{X}^\top(\mathbf{y} - \mathbf{p}) = \mathbf{0}$ no tienen solución cerrada y se resuelven de forma iterativa.
6. La adecuación del modelo se evalúa con la **prueba de razón de verosimilitudes**, comparando el modelo completo contra uno reducido mediante el estadístico (5.6.2), que bajo H_0 es asintóticamente χ_q^2 .